

KAMALI R

B.E. Biomedical Engineering Student

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SUMMARY

Research-oriented Biomedical Engineering student with experience in machine learning, computer vision, biomedical signal processing, and AI-based healthcare systems. Published IEEE conference papers and developed predictive models using Python and OpenCV. I am skilled in data analysis, experimentation, and collaborative research with strong programming abilities and a passion for developing intelligent solutions to real-world problems.

EDUCATION

B.E. Biomedical Engineering, SIMATS Engineering, Saveetha University (2023–2027) **CGPA:** 8.4/10.0

Relevant Coursework: Machine Learning, Medical Imaging, Biomedical Signal Processing, Biomaterials, Biomechanics, Medical Electronics.

PUBLICATIONS

IEEE Publication: Analysis and Prediction of Stroke Using Medium Tree and Coarse Gaussian SVM Classifiers.

ICAMND 2025: Biodegradable Chitosan-Silver Nanocomposite Antibacterial Wound Dressings.

RESEARCH PROJECTS

Eye Disease Detection Using Fundus Images and OpenCV

- Developed an OpenCV-based retinal disease detection system using fundus image processing and feature extraction techniques.

Enhanced Thyroid Anomaly Prediction Using Autoencoder and LightGBM

- Built a machine learning model for thyroid disease prediction using Autoencoder feature extraction and LightGBM classification.

IoT-Based Breath Analysis System for Non-Invasive Diabetes Monitoring

- Designed a Raspberry Pi Pico W-based breath analysis system using gas sensors for non-invasive diabetes monitoring and real-time IoT data transmission.

PROFESSIONAL EXPERIENCE

Biomedical Engineering Intern, Billroth Hospitals (Oct 2025–Nov 2025)

- Calibrated and maintained ECG machines, ventilators, infusion pumps, and patient monitors.
- Conducted preventive maintenance and biomedical safety inspections.

Biomedical Engineering Intern, Kaashiv InfoTech (Aug 2025–Oct 2025)

- Developed biomedical signal processing applications using MATLAB and Python.
- Implemented machine learning models for healthcare data analysis and monitoring systems.

SKILLS

Programming Languages: Python, MATLAB

Tools: Git, GitHub, OpenCV, LabVIEW, ANSYS

Machine Learning: SVM, Random Forest, LightGBM, Deep Learning

Research Areas: Computer Vision, Data Analysis, Signal Processing, Artificial Intelligence

Software: MATLAB, LabVIEW, ANSYS, SolidWorks

CERTIFICATIONS

MATLAB Onramp (MathWorks) | Artificial Intelligence (Simplilearn)